

The price of the second trade P_2 is the initial price plus the impact of the second trade plus the remaining impact from the first trade:

$$P_2 = P_0 + f(x_2) + f(x_1) \cdot e^{-\gamma \cdot 1}$$

The price of the third trade P_3 is the initial price plus the impact of the third trade plus all remaining temporary impact from all previous trades:

$$P_3 = P_0 + f(x_3) + f(x_2) \cdot e^{-\gamma \cdot 1} + f(x_1) \cdot e^{-\gamma \cdot 2}$$

Following, the price of the k^{th} trade P_k is:

$$P_k = P_0 + f(x_k) + f(x_{k-1}) \cdot e^{-\gamma \cdot 1} + \dots + f(x_{k-j}) \cdot e^{-\gamma \cdot j} + \dots + f(x_1) \cdot e^{-\gamma \cdot (k-1)}$$

A general formulation of this expression is:

$$P_k = P_0 + \sum f(x_j) \cdot e^{-0.5 \cdot (k-j)}$$

Illustration 4—Various Market Impact Price Trajectories

Mathematically, market impact is the difference between the price trajectory of the stock with the order and the price trajectory that would have occurred had the order had not been released to or traded in the market. We are not able to observe both price paths simultaneously, only price evolution with the order or price evolution in the absence of the order. Scientists have not figured a way to construct a controlled experiment that will observe both situations simultaneously. Our failure to simultaneously observe both potential price trajectories' market impact has often been described as the Heisenberg uncertainty principle of finance.

Figure 4.4 illustrates four potential effects of market impact cost. Figure 4.4a shows the temporary impact effect of a trade. The buy order pushes the price up and then it reverts to its original path. Figure 4.4b depicts the permanent impact effect of a trade. The buy order pushes the price up. However, after the trade the price does not revert to its original path, but instead is parallel at a level higher than the original path. Figure 4.4c shows a combination of temporary and permanent impact. First, the order pushes the stock price up followed by some temporary reversion, but in this case the price trajectory remains just slightly higher than and parallel to the original trajectory. Figure 4.4d illustrates temporary impact disguised as permanent impact. In this example, the decay of market impact is extremely slow. It is so slow in fact that temporary impact has not